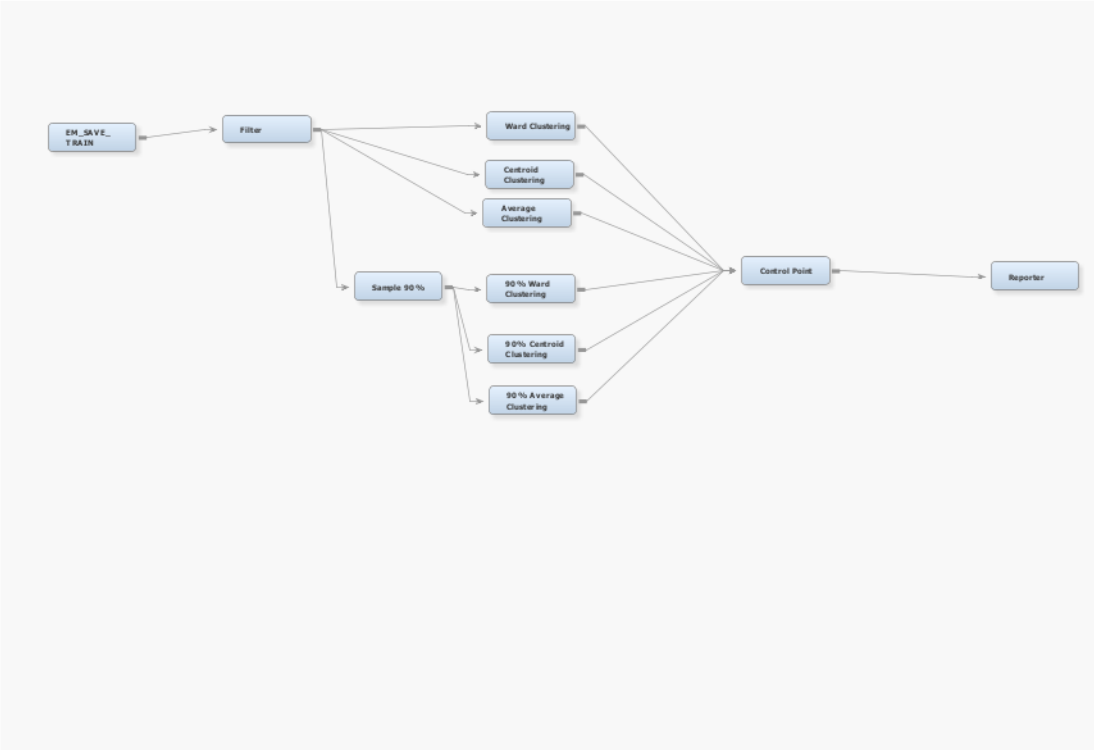
SAS Enterprise Miner Report

User = skunt
Date = 01:27:14 December 09
Project = MC.ClusterAnalysis
Diagram = HU.ClusterAnalysis

Start Node = Report
Node label = Reporter
Nodes = PATH
Showall = N

Format = PDF
Style = LISTING

SAS Enterprise Miner Report
Process Flow Diagram



SAS Enterprise Miner Report

Node=EM_SAVE_TRAIN
Summary

Node id = Ids
Node label = EM_SAVE_TRAIN
Meta path = Ids
Notes =

Node=EM_SAVE_TRAIN
Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	DataSource		DsCreatedBy	skunt		NBytes	1704960	.
ApplyIntervalLevelLowerLimit	Y		DsId	emsavetrain		NCols	10	.
ApplyMaxClassLevels	Y		DsModifiedBy	skunt		NObs	20640	.
ApplyMaxPercentMissing	Y		DsModifyDate	2049326701.3		NewTable		
CMeta	WORK.M28LNUJW		DsSampleName			NewVariableRole	REJECT	
ComputeStatistics	N		DsSampleSize			OutputType	VIEW	
DBPassThrough	Y		DsSampleSizeType			Role	RAW	TRAIN
Data	HUDATA.EM_SAVE_TRAIN		DsScope	LOCAL		Sample	D	
DataSelection	DATASOURCE		IdentifyEmptyColumns	Y		SampleSizeObs	10000	
DataSource	emsavetrain		IntervalLowerLimit	20		SampleSizePercent	20	
DataSourceRole	RAW		Library	HUDATA		SampleSizeType	PERCENT	
Description			MaxClassLevels	20		Scope	LOCAL	
DropMapVariables	Y		MaxPercentMissing	50		Segment		
DsCreateDate	2049326701.2		MetaAdvisor	BASIC		Table	EM_SAVE_TRAIN	

Node=EM_SAVE_TRAIN
Data Attributes

Attribute	Value	Attribute	Value	Attribute	Value
Data Name	EM_SAVE_TRAIN	Date Created	09Dec2024:01:24:06	Data Size	1704960
Data Type	DATA	Date Modified	09Dec2024:01:24:06	Role	RAW
Data Label		Number Rows	20640	Segment	
Engine	BASE	Number Columns	10	Data Library	HUDATA

Node=EM_SAVE_TRAIN
Variables List

Name	Label	Role	Level	Type	Length	Format	Creator
households		INPUT	INTERVAL	N	8	BEST12.0	
housing_median_age		INPUT	INTERVAL	N	8	BEST12.0	
latitude		INPUT	INTERVAL	N	8	BEST12.0	
longitude		INPUT	INTERVAL	N	8	BEST12.0	
median_house_value		INPUT	INTERVAL	N	8	BEST12.0	
median_income		INPUT	INTERVAL	N	8	BEST12.0	
ocean_proximity		INPUT	NOMINAL	C	8	$8.	
population		INPUT	INTERVAL	N	8	BEST12.0	
total_bedrooms		INPUT	INTERVAL	N	8	BEST12.0	
total_rooms		INPUT	INTERVAL	N	8	BEST12.0	

SAS Enterprise Miner Report

Node=Filter
Summary

Node id = Filter
Node label = Filter
Meta path = Ids => Filter
Notes =

Node=Filter
Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	Filter		KeepMissingInterval	Y		PercentsCutoff	0.5	
ClassFilterMethod	MINPCT		MADSCutoff	9		PublishScoreCode	Y	
CreateDistributionData	Y	N	MaxValues	25		SpacingsCutoff	9	
ExportTable	FILTERED		MinFreq	1		StddevCutoff	3	
IntervalFilterMethod	STDDEV		MinPercent	0.01		TabletoFilter	TRAIN	
KeepMissingClass	Y		NormalizeClassValue	Y		UpdateClassLevel	N	

Node=Filter
Variable Summary

Role	Level	Frequency Count	Name
INPUT	INTERVAL	9	households housing_median_age latitude longitude median_house_value median_income population total_bedrooms total_rooms
INPUT	NOMINAL	1	ocean_proximity

Node=Filter
Excluded Class Values

Variable	Role	Level	Train Count	Train Percent	Label	Filter Method
ocean_proximity	INPUT	ISLAND	5	0.024225		MINPCT

Node=Filter
Limits for Interval Variables

Variable	Role	Minimum	Maximum	Filter Method	Keep Missing Values	Label
households	INPUT	-647.45	1646.53	STDDEV	Y	
housing_median_age	INPUT	-9.12	66.40	STDDEV	Y	
latitude	INPUT	29.22	42.04	STDDEV	Y	
longitude	INPUT	-125.58	-113.56	STDDEV	Y	
median_house_value	INPUT	-139331.03	553042.66	STDDEV	Y	
median_income	INPUT	-1.83	9.57	STDDEV	Y	
population	INPUT	-1971.91	4822.86	STDDEV	Y	
total_bedrooms	INPUT	-726.28	1802.03	STDDEV	Y	
total_rooms	INPUT	-3909.08	9180.61	STDDEV	Y	

SAS Enterprise Miner Report

Node=Sample 90%
Summary

Node id = Smpl
Node label = Sample 90%
Meta path = Ids => Filter => Smpl
Notes =

Node=Sample 90%
Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	Sample		IgnoreSmallStrata	N		OutputType	DATA	
AdjustFreq	N		IntervalDistribution	Y		Pvalue	0.01	
Alpha	0.01		LevelProportion	100		RandomSeed	12345	
ClassDistribution	Y		LevelSampleProportion	50		SizeObs	.	
ClusterMethod	RANDOM		LevelSelection	EVENT		SizePercent	90	10
FreqCount	N		Method	DEFAULT		SizeType	PERCENT	
FreqMiss	N		MinStrataSize	5		StratifyCriterion	PROPORTIONAL	

Node=Sample 90%
Variable Summary

Role	Level	Frequency Count	Name
INPUT	INTERVAL	9	households housing_median_age latitude longitude median_house_value median_income population total_bedrooms total_rooms
INPUT	NOMINAL	1	ocean_proximity

SAS Enterprise Miner Report

Node=90% Average Clustering Summary

Node id = Clus6
Node label = 90% Average Clustering
Meta path = Ids => Filter => Smpl => Clus6
Notes =

Node=90% Average Clustering Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	Cluster		FinalMaxNum	20		MissingNominal	DEFAULT	
AutomaticMaxNum	50		HideVariable	Y		MissingOrdinal	DEFAULT	
AutomaticMethod	AVERAGE	WARD	ImputationMethod	NONE		NominalEncoding	GLM	
AutomaticMinNum	2		Initial	DEFAULT		NumberClusterMethod	USER	AUTOMATIC
Bins	100		Learn	.		OrdinalEncoding	RANK	
CCCCutOff	3		LearnFinal	0.02		Radius	0	
ClusterGraphs	Y		LearnInitial	0.5		Stdize	STD	
Clusvar	_SEGMENT_		LearnSteps	1000		Summary	N	
ClusvarLabel	Segment Variable		MaxC	6	10	TrainDefaults	Y	
ClusvarRole	SEGMENT		MaxIter	10		TreeProfile	Y	
DistancePlot	Y		MaxSteps	1200		XConv	0.0001	
Drift	N		MissingInterval	DEFAULT				

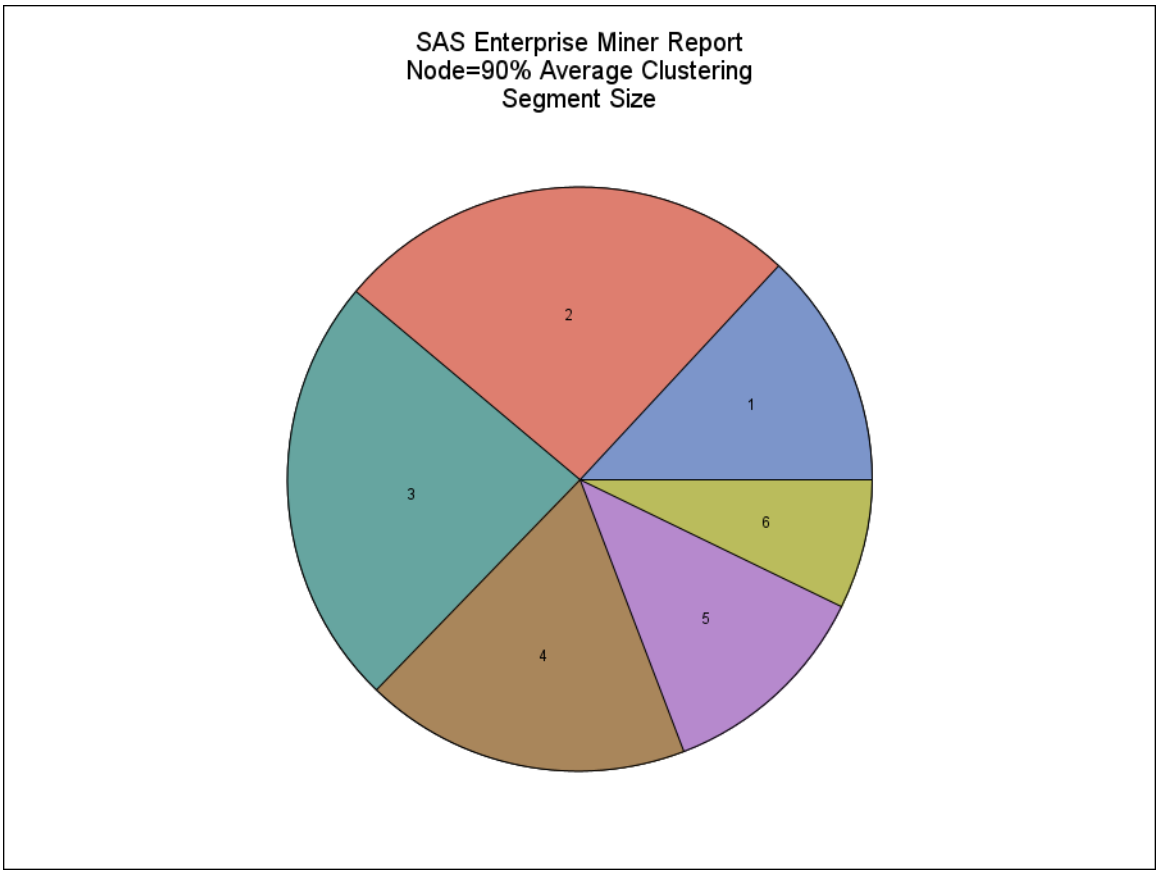
Node=90% Average Clustering Variable Summary

Role	Level	Frequency Count	Name
INPUT	INTERVAL	9	households housing_median_age latitude longitude median_house_value median_income population total_bedrooms total_rooms
INPUT	NOMINAL	1	ocean_proximity
ID	INTERVAL	1	_dataobs_

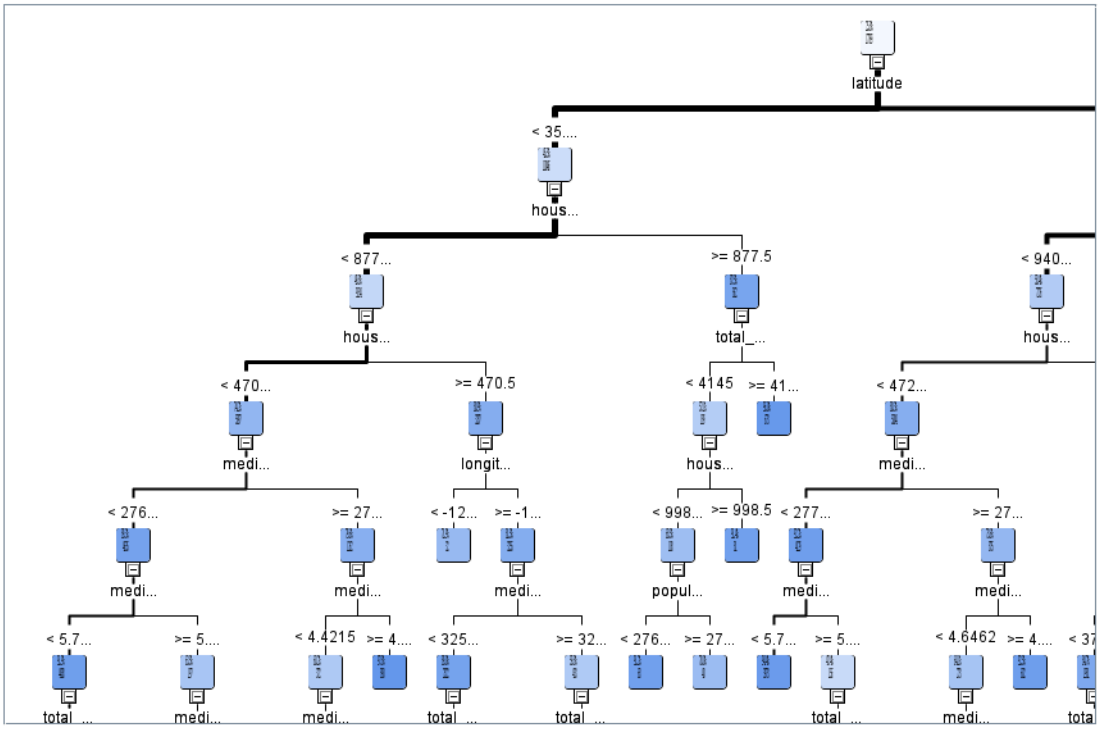
Node=90% Average Clustering Mean Statistics

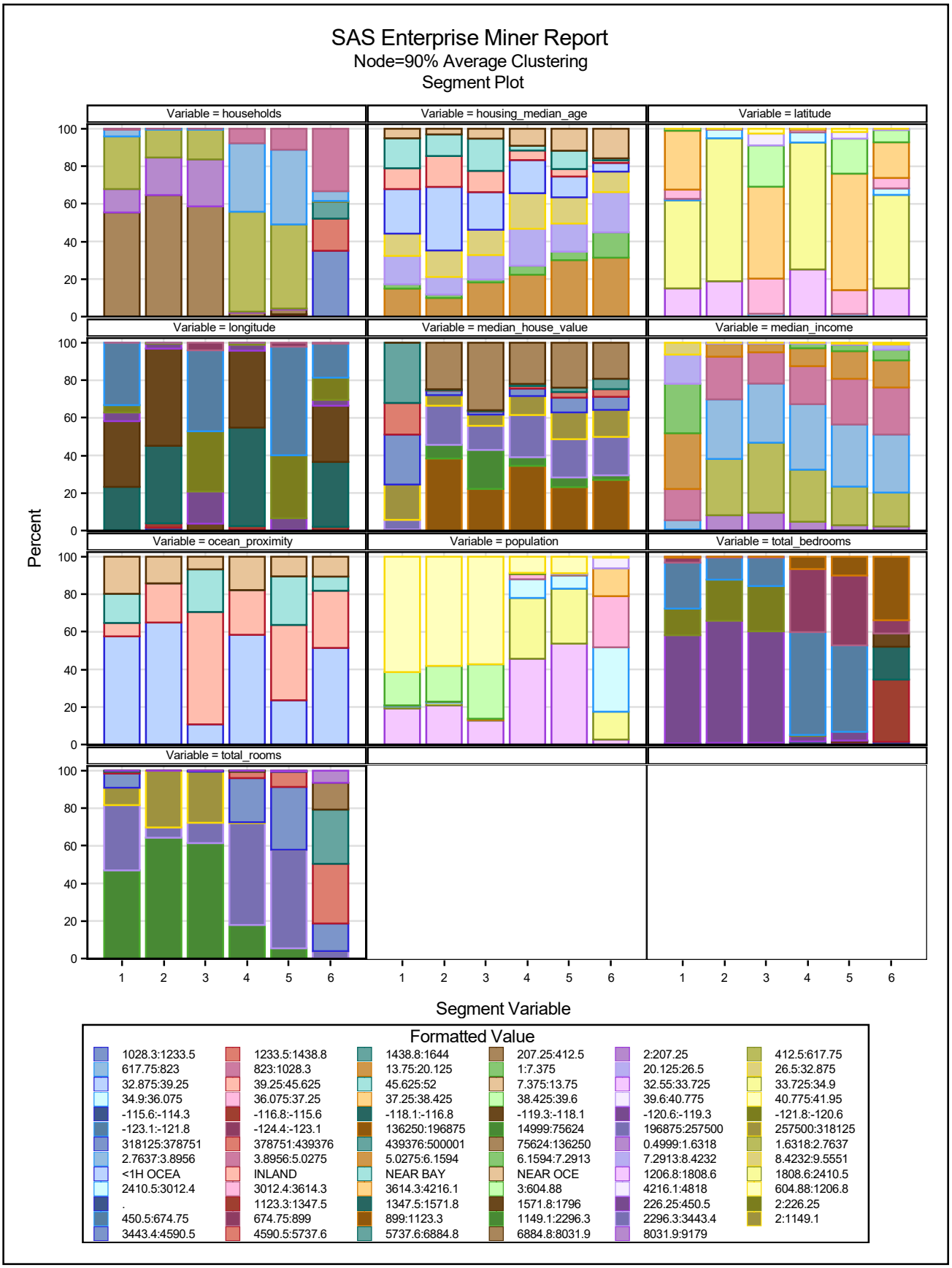
Variable	Segment 1	Segment 2	Segment 3	Segment 4	Segment 5	Segment 6
Clustering Criterion	0.60	0.60	0.60	0.60	0.60	0.60
Maximum Relative Change in Cluster Seeds	0.01	0.01	0.01	0.01	0.01	0.01
Improvement in Clustering Criterion	.	.	.	.	.	.
Frequency of Cluster	2330.00	4576.00	4268.00	3170.00	2150.00	1275.00
Root-Mean-Square Standard Deviation	0.69	0.51	0.58	0.57	0.64	0.77
Maximum Distance from Cluster Seed	5.05	4.27	5.22	4.61	5.24	5.51
Nearest Cluster	2.00	4.00	5.00	2.00	3.00	4.00
Distance to Nearest Cluster	2.97	2.44	2.73	2.44	2.73	3.77
households	356.77	297.68	289.34	616.08	643.12	1110.04
housing_median_age	31.57	33.31	31.45	24.87	24.80	18.64
latitude	35.27	33.90	38.02	33.91	37.94	35.18
longitude	-119.68	-117.97	-121.49	-117.96	-121.81	-119.16
median_house_value	385493.94	168207.34	139881.65	187425.05	204661.73	222122.93

Variable	Segment 1	Segment 2	Segment 3	Segment 4	Segment 5	Segment 6
median_income	6.14	3.23	3.00	3.45	3.81	4.04
population	930.18	934.95	813.45	1826.54	1728.81	3036.49
total_bedrooms	373.41	314.02	316.28	664.60	688.46	1200.36
total_rooms	2231.31	1423.84	1520.42	3018.01	3403.37	5795.77
ocean_proximity=<1H OCEA	0.58	0.65	0.11	0.58	0.24	0.51
ocean_proximity=INLAND	0.07	0.21	0.60	0.24	0.40	0.30
ocean_proximity=NEAR BAY	0.16	-0.00	0.23	0.00	0.26	0.08
ocean_proximity=NEAR OCE	0.20	0.14	0.07	0.18	0.10	0.11



SAS Enterprise Miner Report
Node=90% Average Clustering
Tree Diagram




SAS Enterprise Miner Report
Node=90% Average Clustering
Segment Plot
Variable = households
Variable = housing_median_age
Variable = latitude
Variable = longitude
Variable = median_house_value
Variable = median_income
Variable = ocean_proximity
Variable = population
Variable = total_bedrooms
Variable = total_rooms
Percent
Segment Variable
Formatted Value
1028.3:1233.5
1233.5:1438.8
1438.8:1644
207.25:412.5
2:207.25
412.5:617.75
617.75:823
823:1028.3
13.75:20.125
1:7.375
20.125:26.5
26.5:32.875
32.875:39.25
39.25:45.625
45.625:52
7.375:13.75
32.55:33.725
33.725:34.9
34.9:36.075
36.075:37.25
37.25:38.425
38.425:39.6
39.6:40.775
40.775:41.95
-115.6:-114.3
-116.8:-115.6
-118.1:-116.8
-119.3:-118.1
-120.6:-119.3
-121.8:-120.6
-123.1:-121.8
-124.4:-123.1
136250:196875
14999:75624
196875:257500
257500:318125
318125:378751
378751:439376
439376:500001
75624:136250
0.4999:1.6318
1.6318:2.7637
2.7637:3.8956
3.8956:5.0275
5.0275:6.1594
6.1594:7.2913
7.2913:8.4232
8.4232:9.5551
<1H OCEA
INLAND
NEAR BAY
NEAR OCE
1206.8:1808.6
1808.6:2410.5
2410.5:3012.4
3012.4:3614.3
3614.3:4216.1
3:604.88
4216.1:4818
604.88:1206.8
.
1123.3:1347.5
1347.5:1571.8
1571.8:1796
226.25:450.5
2:226.25
450.5:674.75
674.75:899
899:1123.3
1149.1:2296.3
2296.3:3443.4
2:1149.1
3443.4:4590.5
4590.5:5737.6
5737.6:6884.8
6884.8:8031.9
8031.9:9179

SAS Enterprise Miner Report

Node=90% Centroid Clustering Summary

Node id = Clus5
Node label = 90% Centroid Clustering
Meta path = Ids => Filter => Smpl => Clus5
Notes =

Node=90% Centroid Clustering Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	Cluster		FinalMaxNum	20		MissingNominal	DEFAULT	
AutomaticMaxNum	50		HideVariable	Y		MissingOrdinal	DEFAULT	
AutomaticMethod	CENTROID	WARD	ImputationMethod	NONE		NominalEncoding	GLM	
AutomaticMinNum	2		Initial	DEFAULT		NumberClusterMethod	USER	AUTOMATIC
Bins	100		Learn	.		OrdinalEncoding	RANK	
CCCCutOff	3		LearnFinal	0.02		Radius	0	
ClusterGraphs	Y		LearnInitial	0.5		Stdize	STD	
Clusvar	_SEGMENT_		LearnSteps	1000		Summary	N	
ClusvarLabel	Segment Variable		MaxC	6	10	TrainDefaults	Y	
ClusvarRole	SEGMENT		MaxIter	10		TreeProfile	Y	
DistancePlot	Y		MaxSteps	1200		XConv	0.0001	
Drift	N		MissingInterval	DEFAULT				

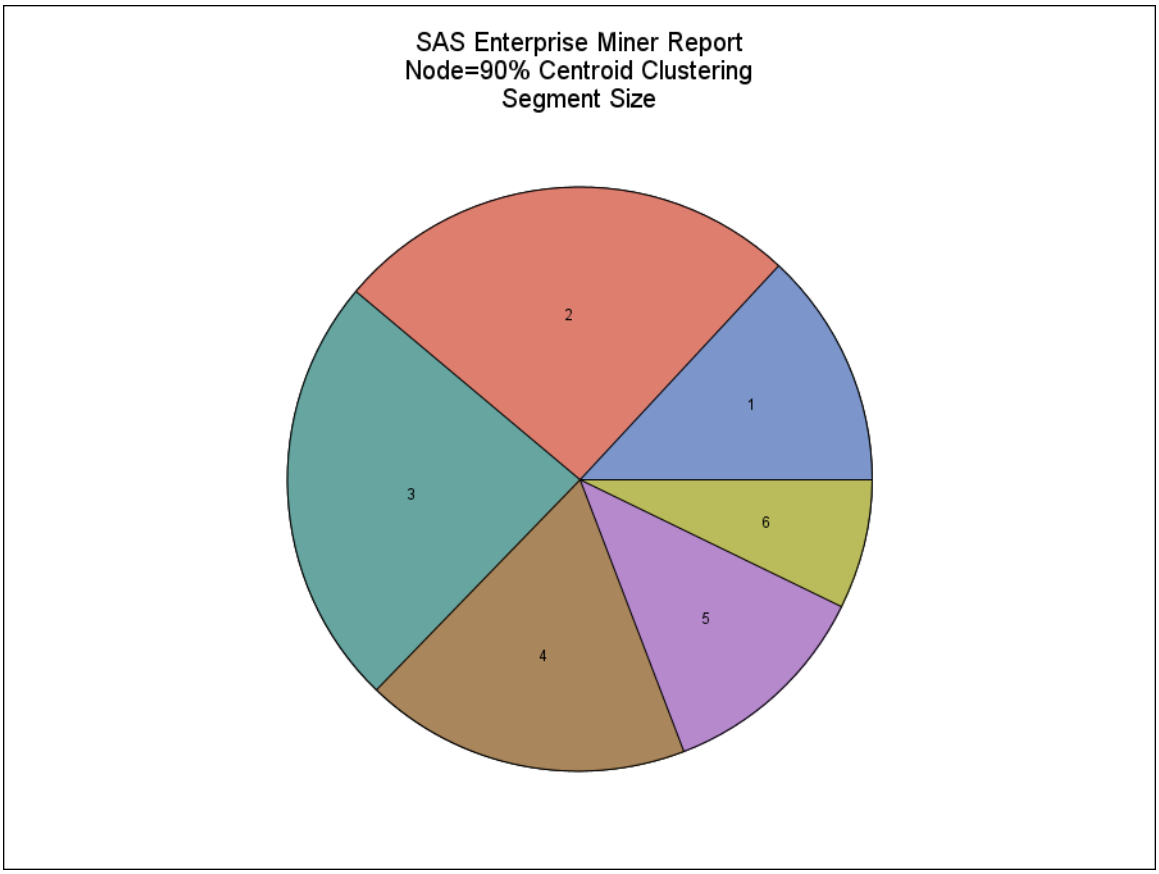
Node=90% Centroid Clustering Variable Summary

Role	Level	Frequency Count	Name
INPUT	INTERVAL	9	households housing_median_age latitude longitude median_house_value median_income population total_bedrooms total_rooms
INPUT	NOMINAL	1	ocean_proximity
ID	INTERVAL	1	_dataobs_

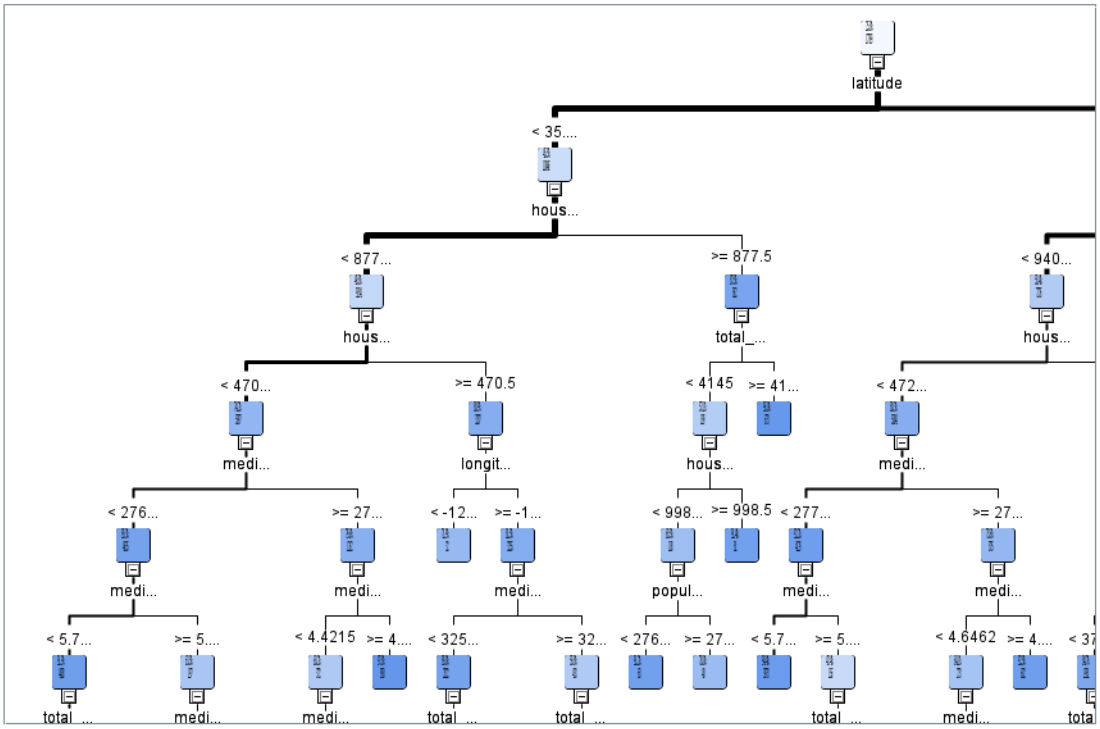
Node=90% Centroid Clustering Mean Statistics

Variable	Segment 1	Segment 2	Segment 3	Segment 4	Segment 5	Segment 6
Clustering Criterion	0.60	0.60	0.60	0.60	0.60	0.60
Maximum Relative Change in Cluster Seeds	0.01	0.01	0.01	0.01	0.01	0.01
Improvement in Clustering Criterion	.	.	.	.	.	.
Frequency of Cluster	2330.00	4576.00	4268.00	3170.00	2150.00	1275.00
Root-Mean-Square Standard Deviation	0.69	0.51	0.58	0.57	0.64	0.77
Maximum Distance from Cluster Seed	5.05	4.27	5.22	4.61	5.24	5.51
Nearest Cluster	2.00	4.00	5.00	2.00	3.00	4.00
Distance to Nearest Cluster	2.97	2.44	2.73	2.44	2.73	3.77
households	356.77	297.68	289.34	616.08	643.12	1110.04
housing_median_age	31.57	33.31	31.45	24.87	24.80	18.64
latitude	35.27	33.90	38.02	33.91	37.94	35.18
longitude	-119.68	-117.97	-121.49	-117.96	-121.81	-119.16
median_house_value	385493.94	168207.34	139881.65	187425.05	204661.73	222122.93

Variable	Segment 1	Segment 2	Segment 3	Segment 4	Segment 5	Segment 6
median_income	6.14	3.23	3.00	3.45	3.81	4.04
population	930.18	934.95	813.45	1826.54	1728.81	3036.49
total_bedrooms	373.41	314.02	316.28	664.60	688.46	1200.36
total_rooms	2231.31	1423.84	1520.42	3018.01	3403.37	5795.77
ocean_proximity=<1H OCEA	0.58	0.65	0.11	0.58	0.24	0.51
ocean_proximity=INLAND	0.07	0.21	0.60	0.24	0.40	0.30
ocean_proximity=NEAR BAY	0.16	-0.00	0.23	0.00	0.26	0.08
ocean_proximity=NEAR OCE	0.20	0.14	0.07	0.18	0.10	0.11



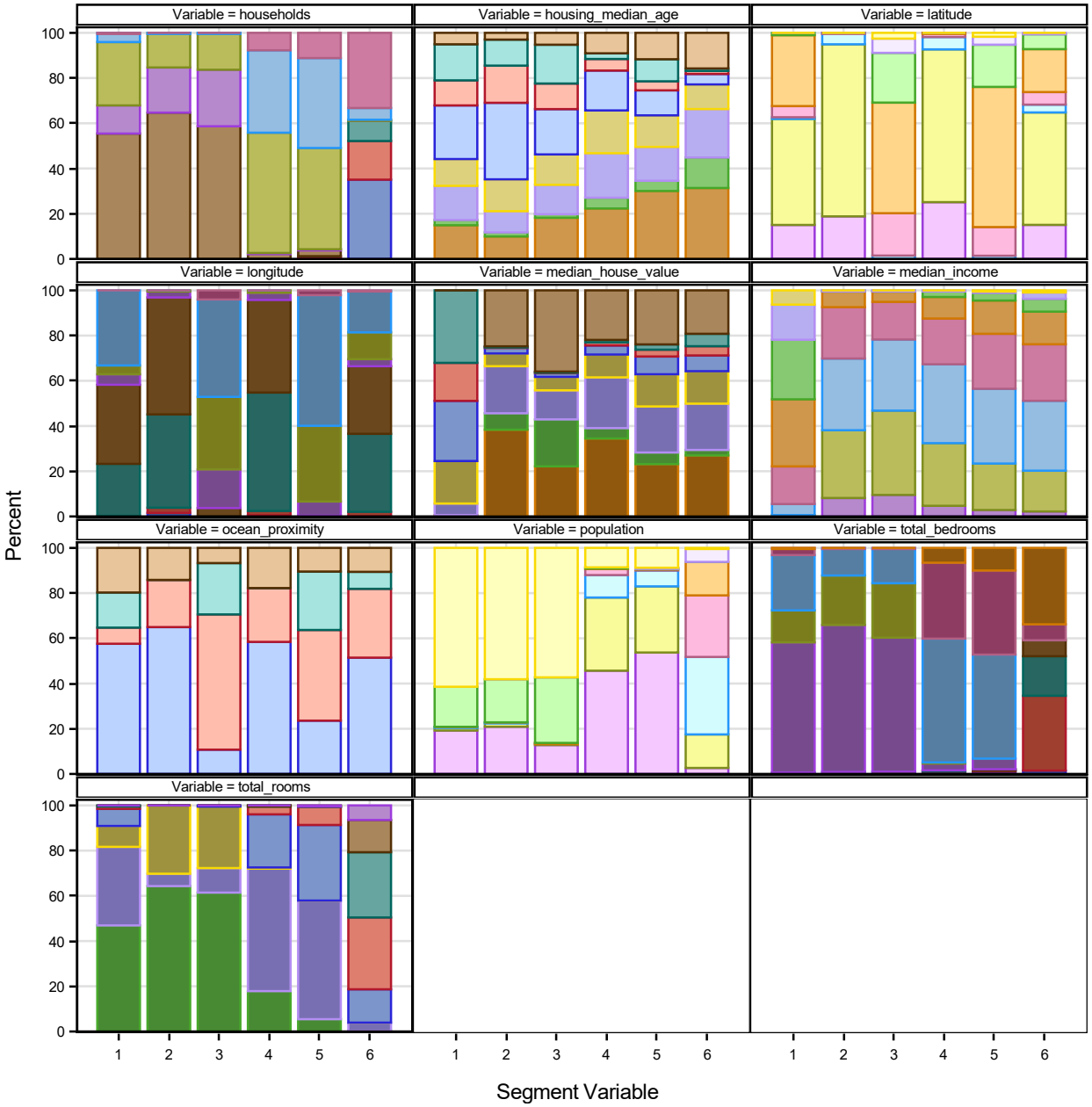
SAS Enterprise Miner Report
Node=90% Centroid Clustering
Tree Diagram



SAS Enterprise Miner Report

Node=90% Centroid Clustering

Segment Plot



Formatted Value

1028.3:1233.5	1233.5:1438.8	1438.8:1644	207.25:412.5	2:207.25	412.5:617.75
617.75:823	823:1028.3	13.75:20.125	1:7.375	20.125:26.5	26.5:32.875
32.875:39.25	39.25:45.625	45.625:52	7.375:13.75	32.55:33.725	33.725:34.9
34.9:36.075	36.075:37.25	37.25:38.425	38.425:39.6	39.6:40.775	40.775:41.95
-115.6:-114.3	-116.8:-115.6	-118.1:-116.8	-119.3:-118.1	-120.6:-119.3	-121.8:-120.6
-123.1:-121.8	-124.4:-123.1	136250:196875	14999:75624	196875:257500	257500:318125
318125:378751	378751:439376	439376:500001	75624:136250	0.4999:1.6318	1.6318:2.7637
2.7637:3.8956	3.8956:5.0275	5.0275:6.1594	6.1594:7.2913	7.2913:8.4232	8.4232:9.5551
<1H OCEA	INLAND	NEAR BAY	NEAR OCE	1206.8:1808.6	1808.6:2410.5
2410.5:3012.4	3012.4:3614.3	3614.3:4216.1	3:604.88	4216.1:4818	604.88:1206.8
.	1123.3:1347.5	1347.5:1571.8	1571.8:1796	226.25:450.5	2:226.25
450.5:674.75	674.75:899	899:1123.3	1149.1:2296.3	2296.3:3443.4	2:1149.1
3443.4:4590.5	4590.5:5737.6	5737.6:6884.8	6884.8:8031.9	8031.9:9179	

SAS Enterprise Miner Report

Node=90% Ward Clustering Summary

Node id = Clus2
Node label = 90% Ward Clustering
Meta path = Ids => Filter => Smpl => Clus2
Notes =

Node=90% Ward Clustering Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	Cluster		FinalMaxNum	20		MissingNominal	DEFAULT	
AutomaticMaxNum	50		HideVariable	Y		MissingOrdinal	DEFAULT	
AutomaticMethod	WARD		ImputationMethod	NONE		NominalEncoding	GLM	
AutomaticMinNum	2		Initial	DEFAULT		NumberClusterMethod	AUTOMATIC	
Bins	100		Learn	.		OrdinalEncoding	RANK	
CCCCutOff	3		LearnFinal	0.02		Radius	0	
ClusterGraphs	Y		LearnInitial	0.5		Stdize	STD	
Clusvar	_SEGMENT_		LearnSteps	1000		Summary	N	
ClusvarLabel	Segment Variable		MaxC	10		TrainDefaults	Y	
ClusvarRole	SEGMENT		MaxIter	10		TreeProfile	Y	
DistancePlot	Y		MaxSteps	1200		XConv	0.0001	
Drift	N		MissingInterval	DEFAULT				

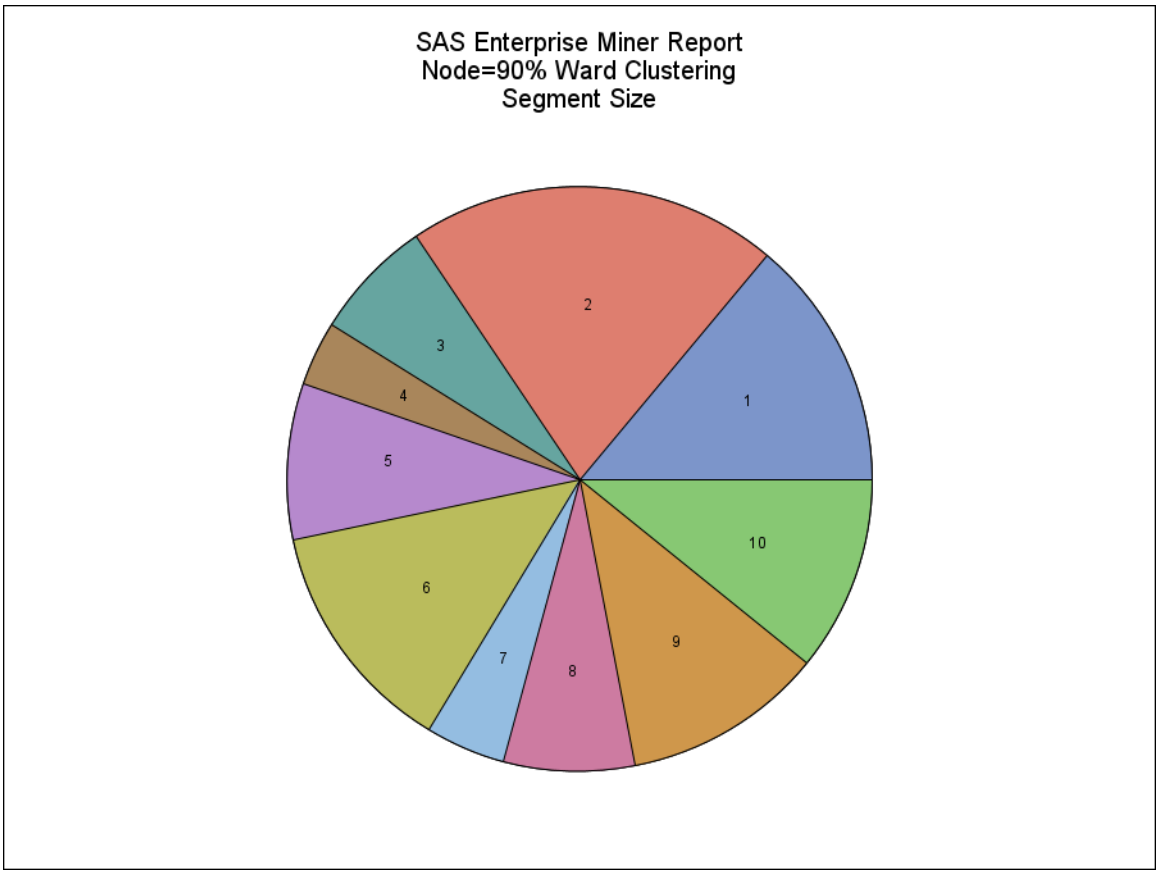
Node=90% Ward Clustering Variable Summary

Role	Level	Frequency Count	Name
INPUT	INTERVAL	9	households housing_median_age latitude longitude median_house_value median_income population total_bedrooms total_rooms
INPUT	NOMINAL	1	ocean_proximity
ID	INTERVAL	1	_dataobs_

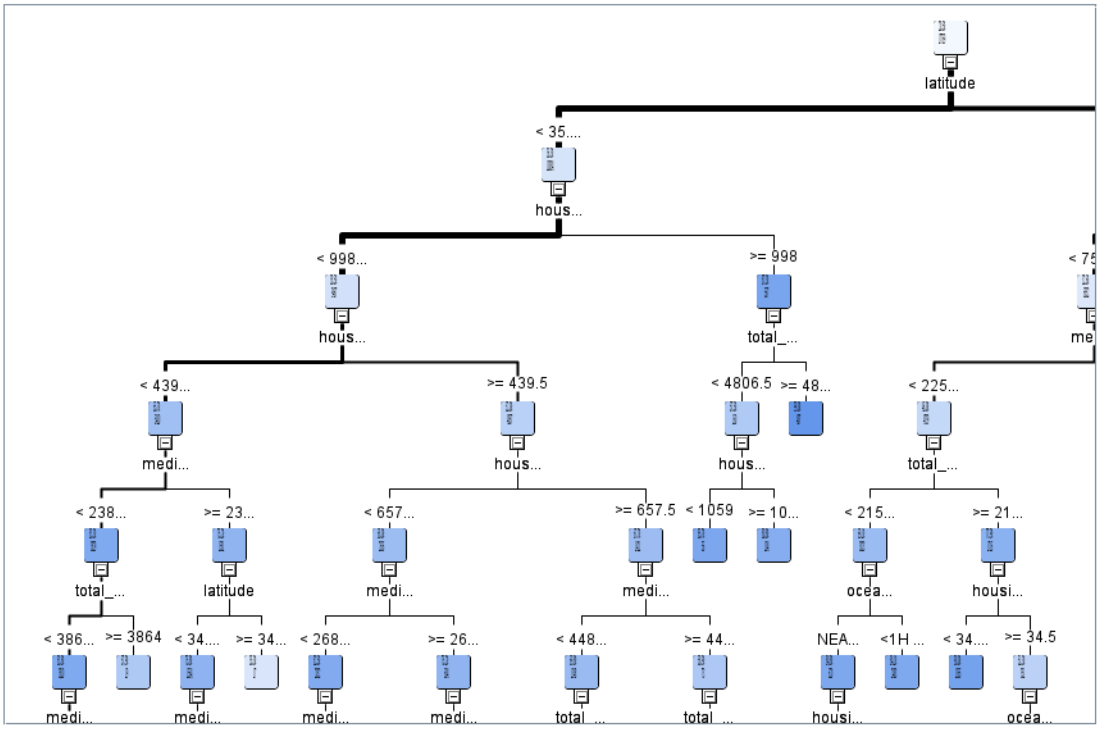
Node=90% Ward Clustering Mean Statistics

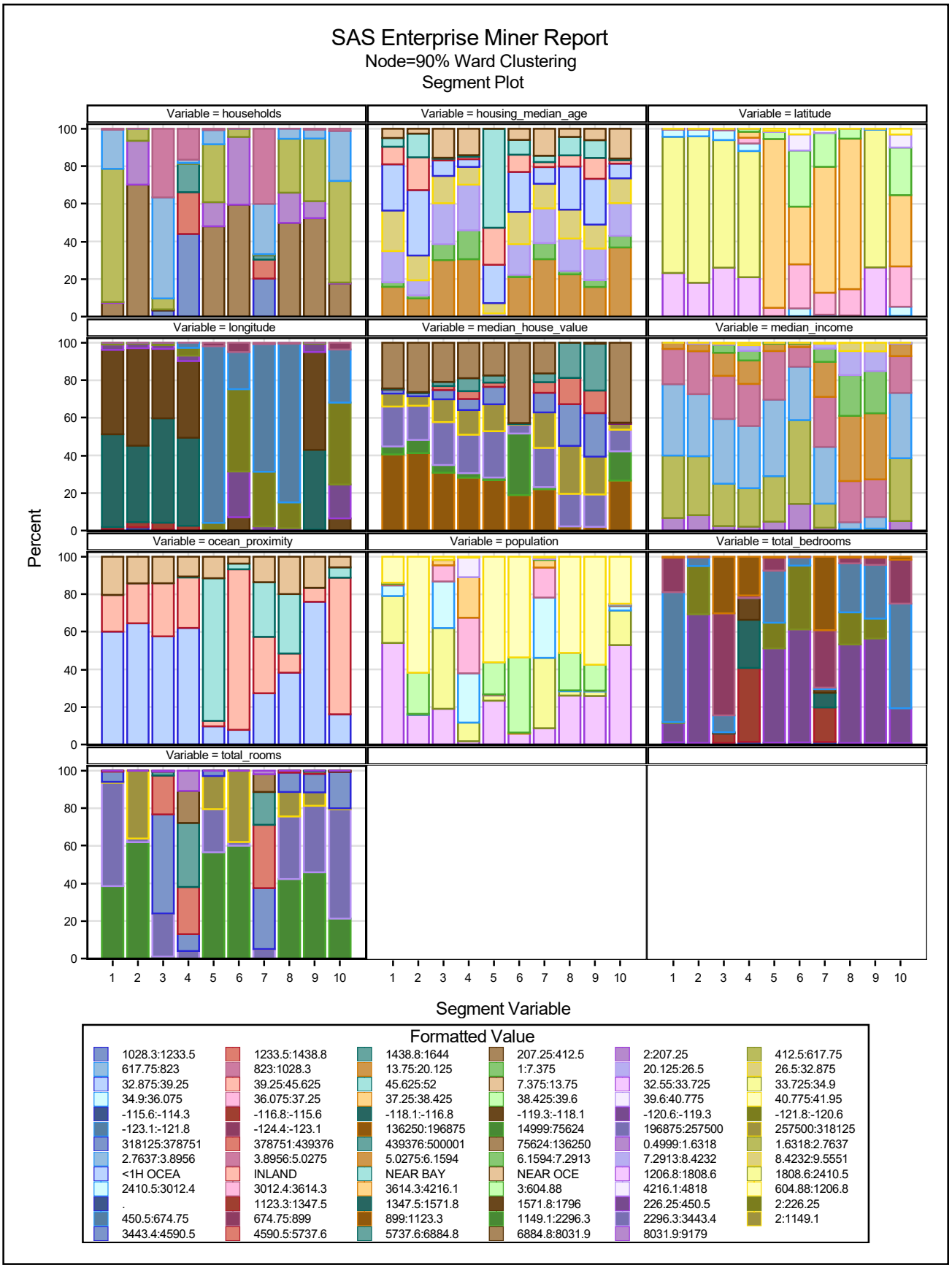
Variable	Segment 1	Segment 2	Segment 3	Segment 4	Segment 5	Segment 6	Segment 7	Segment 8	Segment 9	Segment 10
Clustering Criterion	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54
Maximum Relative Change in Cluster Seeds	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04
Improvement in Clustering Criterion	.	.	.	.	.	.	.	.	.	.
Frequency of Cluster	2479.00	3648.00	1198.00	625.00	1537.00	2296.00	795.00	1293.00	1997.00	1901.00
Root-Mean-Square Standard Deviation	0.50	0.49	0.57	0.71	0.53	0.49	0.70	0.62	0.58	0.53
Maximum Distance from Cluster Seed	4.26	4.06	4.60	5.47	4.12	5.28	5.16	4.02	4.84	4.77
Nearest Cluster	2.00	1.00	1.00	3.00	8.00	10.00	3.00	5.00	2.00	6.00
Distance to Nearest Cluster	1.98	1.98	2.09	2.89	2.49	2.32	3.07	2.49	2.62	2.32
households	540.90	274.62	792.84	1194.16	381.19	239.66	966.16	358.50	378.86	534.46
housing_median_age	28.65	33.99	20.07	18.40	44.67	28.65	21.33	28.77	29.41	20.13
latitude	33.85	33.89	33.91	34.22	37.81	38.06	37.84	37.58	33.87	37.96
longitude	-117.95	-117.94	-117.88	-118.21	-122.28	-121.11	-121.91	-122.07	-118.16	-121.25

Variable	Segment 1	Segment 2	Segment 3	Segment 4	Segment 5	Segment 6	Segment 7	Segment 8	Segment 9	Segment 10
median_house_value	172951.43	163220.59	198894.84	223639.24	220708.42	106260.10	237722.41	342755.88	355046.14	134493.32
median_income	3.09	3.15	3.74	3.96	3.33	2.67	4.25	5.92	5.83	3.23
population	1657.56	869.19	2279.06	3263.88	990.38	685.47	2542.82	974.80	1008.47	1501.30
total_bedrooms	579.54	289.64	865.32	1297.03	403.52	267.16	1029.08	372.12	396.52	581.69
total_rooms	2493.06	1304.62	4037.27	6145.35	1844.18	1277.67	5182.75	2232.63	2322.93	2865.08
ocean_proximity=<1H OCEA	0.60	0.65	0.58	0.62	0.10	0.08	0.27	0.38	0.76	0.16
ocean_proximity=INLAND	0.20	0.21	0.28	0.27	0.03	0.85	0.30	0.10	0.07	0.73
ocean_proximity=NEAR BAY	0.00	0.00	-0.00	0.00	0.76	0.03	0.29	0.32	0.00	0.05
ocean_proximity=NEAR OCE	0.20	0.14	0.14	0.11	0.11	0.04	0.14	0.20	0.17	0.06



SAS Enterprise Miner Report
Node=90% Ward Clustering
Tree Diagram




SAS Enterprise Miner Report
Node=90% Ward Clustering
Segment Plot
Variable = households
Variable = housing_median_age
Variable = latitude
Variable = longitude
Variable = median_house_value
Variable = median_income
Variable = ocean_proximity
Variable = population
Variable = total_bedrooms
Variable = total_rooms
Percent
Segment Variable
Formatted Value
1028.3:1233.5
1233.5:1438.8
1438.8:1644
207.25:412.5
2:207.25
412.5:617.75
617.75:823
823:1028.3
13.75:20.125
1:7.375
20.125:26.5
26.5:32.875
32.875:39.25
39.25:45.625
45.625:52
7.375:13.75
32.55:33.725
33.725:34.9
34.9:36.075
36.075:37.25
37.25:38.425
38.425:39.6
39.6:40.775
40.775:41.95
-115.6:-114.3
-116.8:-115.6
-118.1:-116.8
-119.3:-118.1
-120.6:-119.3
-121.8:-120.6
-123.1:-121.8
-124.4:-123.1
136250:196875
14999:75624
196875:257500
257500:318125
318125:378751
378751:439376
439376:500001
75624:136250
0.4999:1.6318
1.6318:2.7637
2.7637:3.8956
3.8956:5.0275
5.0275:6.1594
6.1594:7.2913
7.2913:8.4232
8.4232:9.5551
<1H OCEA
INLAND
NEAR BAY
NEAR OCE
1206.8:1808.6
1808.6:2410.5
2410.5:3012.4
3012.4:3614.3
3614.3:4216.1
3:604.88
4216.1:4818
604.88:1206.8
.
1123.3:1347.5
1347.5:1571.8
1571.8:1796
226.25:450.5
2:226.25
450.5:674.75
674.75:899
899:1123.3
1149.1:2296.3
2296.3:3443.4
2:1149.1
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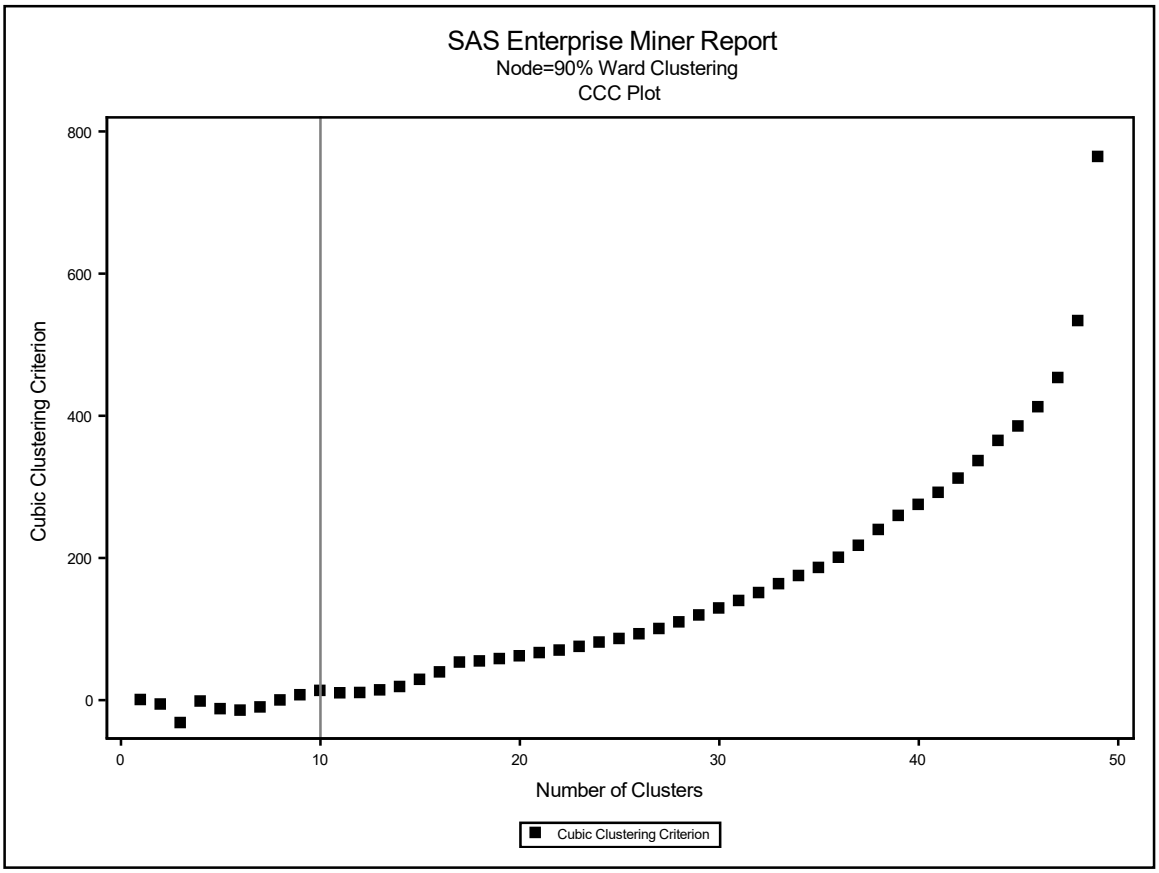
4590.5:5737.6
5737.6:6884.8
6884.8:8031.9
8031.9:9179


SAS Enterprise Miner Report
Node=90% Ward Clustering
CCC Plot
Cubic Clustering Criterion
800
600
400
200
0
0
10
20
30
40
50
Number of Clusters
Cubic Clustering Criterion

SAS Enterprise Miner Report

Node=Average Clustering Summary

Node id = Clus4
Node label = Average Clustering
Meta path = Ids => Filter => Clus4
Notes =

Node=Average Clustering Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	Cluster		FinalMaxNum	20		MissingNominal	DEFAULT	
AutomaticMaxNum	50		HideVariable	Y		MissingOrdinal	DEFAULT	
AutomaticMethod	AVERAGE	WARD	ImputationMethod	NONE		NominalEncoding	GLM	
AutomaticMinNum	2		Initial	DEFAULT		NumberClusterMethod	USER	AUTOMATIC
Bins	100		Learn	.		OrdinalEncoding	RANK	
CCCCutOff	3		LearnFinal	0.02		Radius	0	
ClusterGraphs	Y		LearnInitial	0.5		Stdize	STD	
Clusvar	_SEGMENT_		LearnSteps	1000		Summary	N	
ClusvarLabel	Segment Variable		MaxC	6	10	TrainDefaults	Y	
ClusvarRole	SEGMENT		MaxIter	10		TreeProfile	Y	
DistancePlot	Y		MaxSteps	1200		XConv	0.0001	
Drift	N		MissingInterval	DEFAULT				

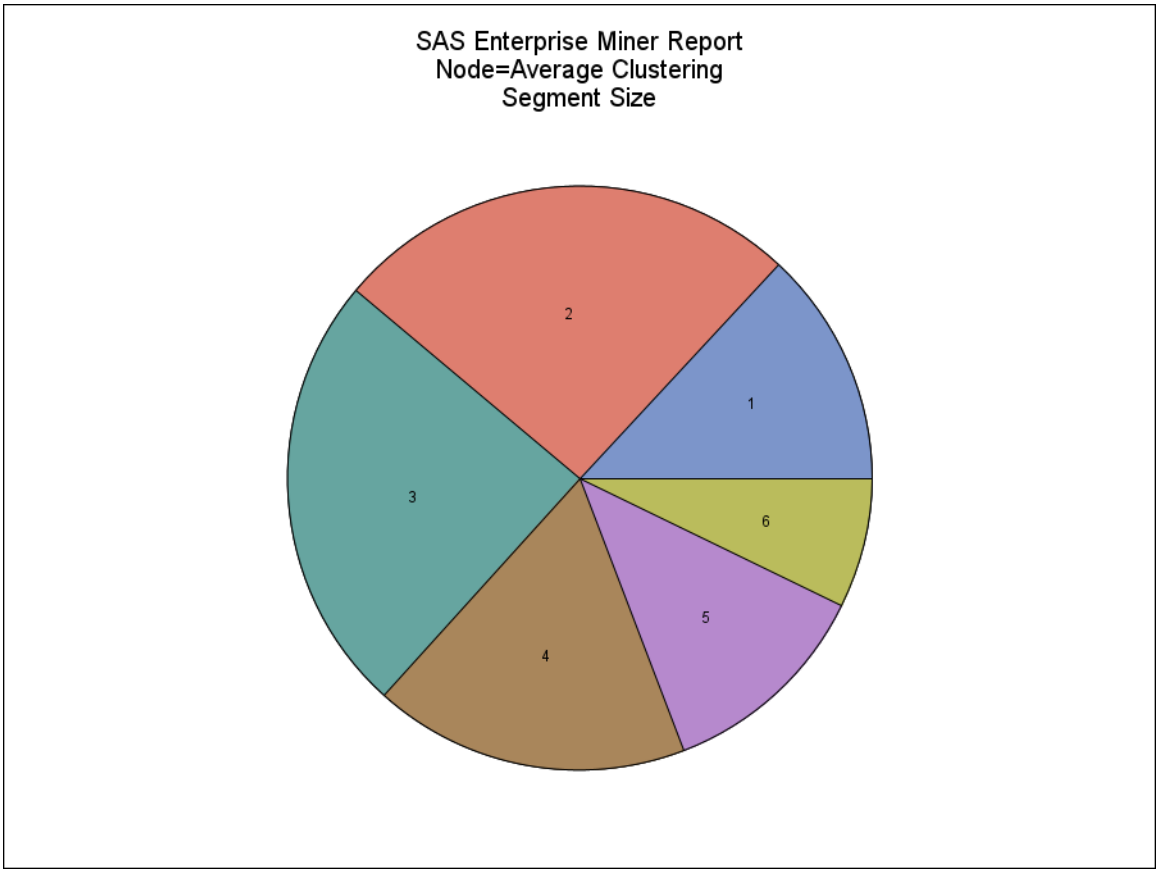
Node=Average Clustering Variable Summary

Role	Level	Frequency Count	Name
INPUT	INTERVAL	9	households housing_median_age latitude longitude median_house_value median_income population total_bedrooms total_rooms
INPUT	NOMINAL	1	ocean_proximity

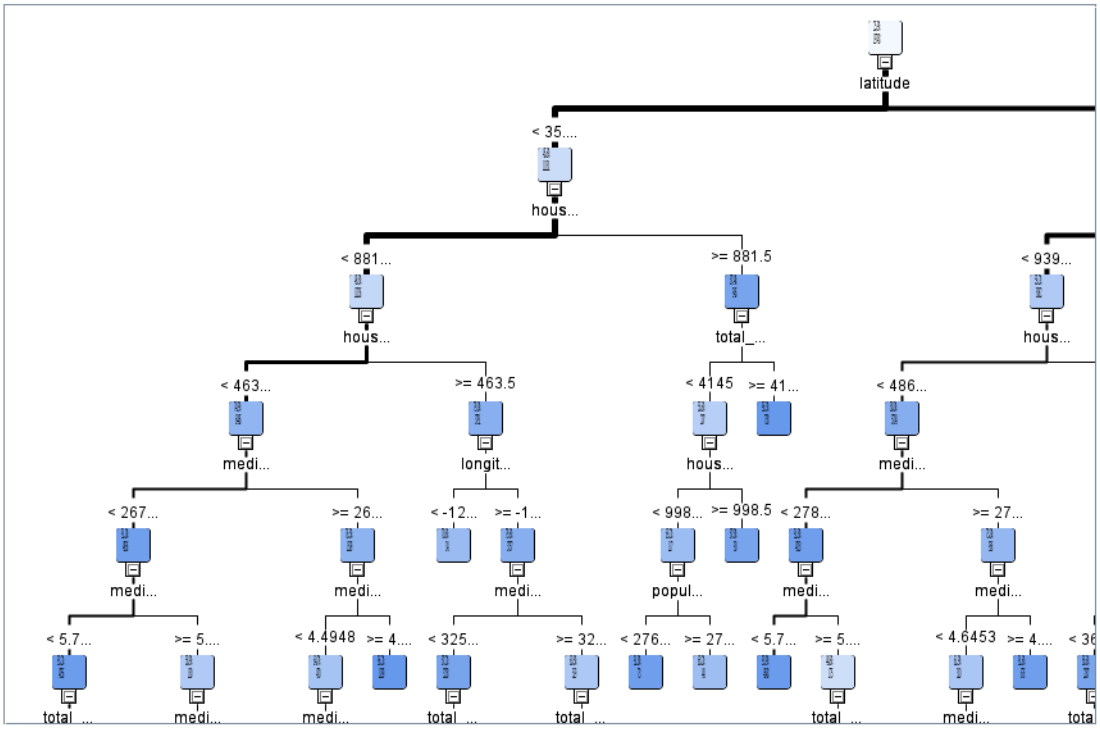
Node=Average Clustering Mean Statistics

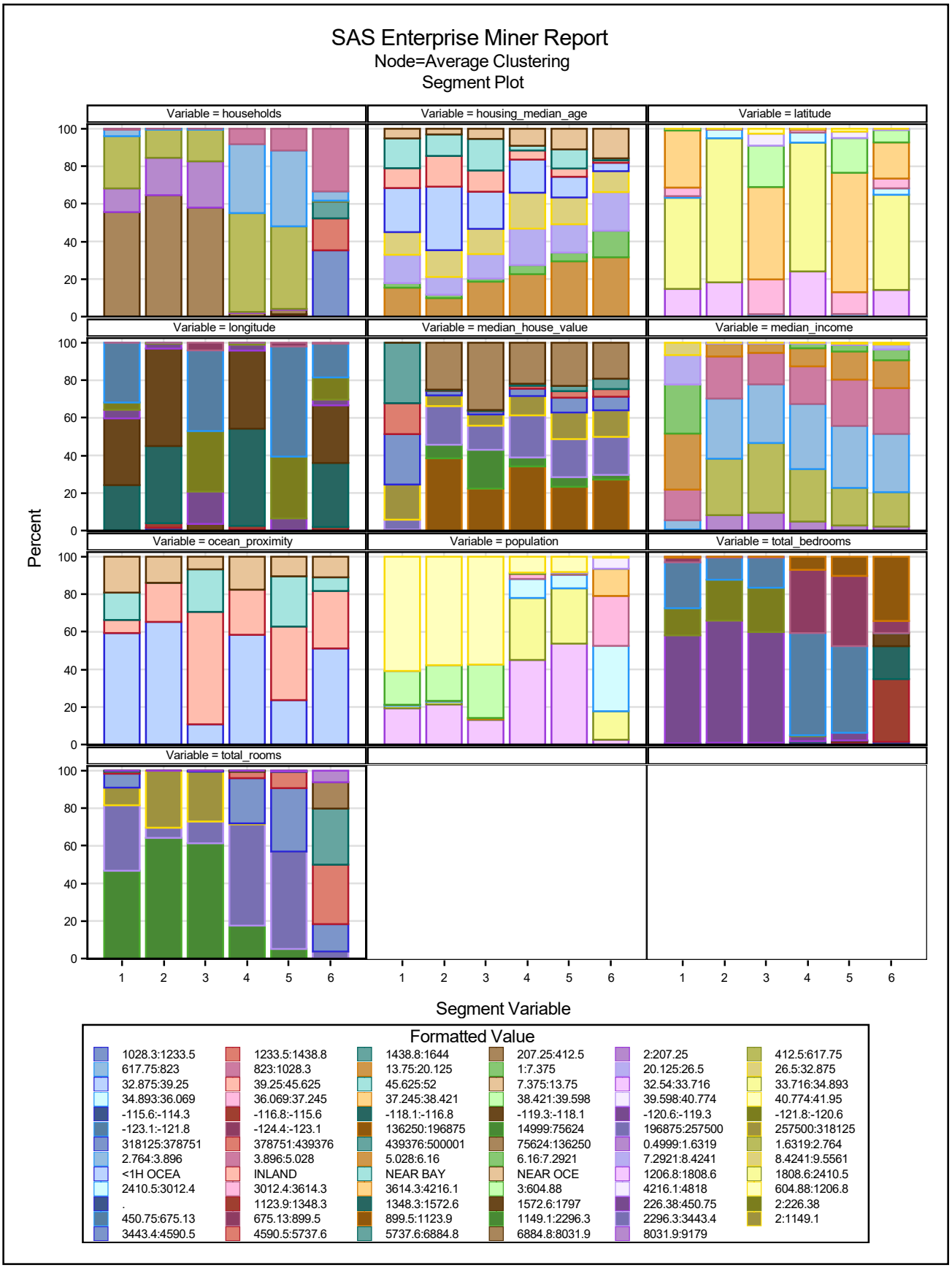
Variable	Segment 1	Segment 2	Segment 3	Segment 4	Segment 5	Segment 6
Clustering Criterion	0.60	0.60	0.60	0.60	0.60	0.60
Maximum Relative Change in Cluster Seeds	0.01	0.01	0.01	0.01	0.01	0.01
Improvement in Clustering Criterion	.	.	.	.	.	.
Frequency of Cluster	2586.00	5106.00	4792.00	3497.00	2359.00	1403.00
Root-Mean-Square Standard Deviation	0.69	0.50	0.58	0.57	0.64	0.77
Maximum Distance from Cluster Seed	5.04	4.27	5.22	4.61	5.25	5.51
Nearest Cluster	2.00	4.00	5.00	2.00	3.00	4.00
Distance to Nearest Cluster	2.96	2.46	2.74	2.46	2.74	3.75
households	355.69	298.41	292.30	619.66	646.59	1110.82
housing_median_age	31.36	33.30	31.26	24.78	25.13	18.50
latitude	35.22	33.90	38.03	33.92	37.93	35.18
longitude	-119.62	-117.97	-121.50	-117.96	-121.82	-119.16
median_house_value	385426.55	167948.69	140105.13	187661.23	207806.97	222046.29
median_income	6.15	3.22	3.01	3.45	3.83	4.03

Variable	Segment 1	Segment 2	Segment 3	Segment 4	Segment 5	Segment 6
population	930.74	938.88	821.70	1832.69	1739.93	3033.01
total_bedrooms	371.76	314.66	319.45	668.87	692.78	1201.51
total_rooms	2230.01	1424.32	1537.28	3031.05	3428.03	5793.53
ocean_proximity=<1H OCEA	0.59	0.65	0.11	0.58	0.24	0.51
ocean_proximity=INLAND	0.07	0.21	0.60	0.24	0.39	0.31
ocean_proximity=NEAR BAY	0.15	0.00	0.23	0.00	0.27	0.07
ocean_proximity=NEAR OCE	0.19	0.14	0.07	0.18	0.10	0.11



SAS Enterprise Miner Report
Node=Average Clustering
Tree Diagram




SAS Enterprise Miner Report
Node=Average Clustering
Segment Plot
Variable = households
Variable = housing_median_age
Variable = latitude
Variable = longitude
Variable = median_house_value
Variable = median_income
Variable = ocean_proximity
Variable = population
Variable = total_bedrooms
Variable = total_rooms
Percent
Segment Variable
Formatted Value
1028.3:1233.5
1233.5:1438.8
1438.8:1644
207.25:412.5
2:207.25
412.5:617.75
617.75:823
823:1028.3
13.75:20.125
1:7.375
20.125:26.5
26.5:32.875
32.875:39.25
39.25:45.625
45.625:52
7.375:13.75
32.54:33.716
33.716:34.893
34.893:36.069
36.069:37.245
37.245:38.421
38.421:39.598
39.598:40.774
40.774:41.95
-115.6:-114.3
-116.8:-115.6
-118.1:-116.8
-119.3:-118.1
-120.6:-119.3
-121.8:-120.6
-123.1:-121.8
-124.4:-123.1
136250:196875
14999:75624
196875:257500
257500:318125
318125:378751
378751:439376
439376:500001
75624:136250
0.4999:1.6319
1.6319:2.764
2.764:3.896
3.896:5.028
5.028:6.16
6.16:7.2921
7.2921:8.4241
8.4241:9.5561
<1H OCEA
INLAND
NEAR BAY
NEAR OCE
1206.8:1808.6
1808.6:2410.5
2410.5:3012.4
3012.4:3614.3
3614.3:4216.1
3:604.88
4216.1:4818
604.88:1206.8
.
1123.9:1348.3
1348.3:1572.6
1572.6:1797
226.38:450.75
2:226.38
450.75:675.13
675.13:899.5
899.5:1123.9
1149.1:2296.3
2296.3:3443.4
2:1149.1
3443.4:4590.5
4590.5:5737.6
5737.6:6884.8
6884.8:8031.9
8031.9:9179

SAS Enterprise Miner Report

Node=Centroid Clustering Summary

Node id = Clus3
Node label = Centroid Clustering
Meta path = Ids => Filter => Clus3
Notes =

Node=Centroid Clustering Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	Cluster		FinalMaxNum	20		MissingNominal	DEFAULT	
AutomaticMaxNum	50		HideVariable	Y		MissingOrdinal	DEFAULT	
AutomaticMethod	CENTROID	WARD	ImputationMethod	NONE		NominalEncoding	GLM	
AutomaticMinNum	2		Initial	DEFAULT		NumberClusterMethod	USER	AUTOMATIC
Bins	100		Learn	.		OrdinalEncoding	RANK	
CCCCutOff	3		LearnFinal	0.02		Radius	0	
ClusterGraphs	Y		LearnInitial	0.5		Stdize	STD	
Clusvar	_SEGMENT_		LearnSteps	1000		Summary	N	
ClusvarLabel	Segment Variable		MaxC	6	10	TrainDefaults	Y	
ClusvarRole	SEGMENT		MaxIter	10		TreeProfile	Y	
DistancePlot	Y		MaxSteps	1200		XConv	0.0001	
Drift	N		MissingInterval	DEFAULT				

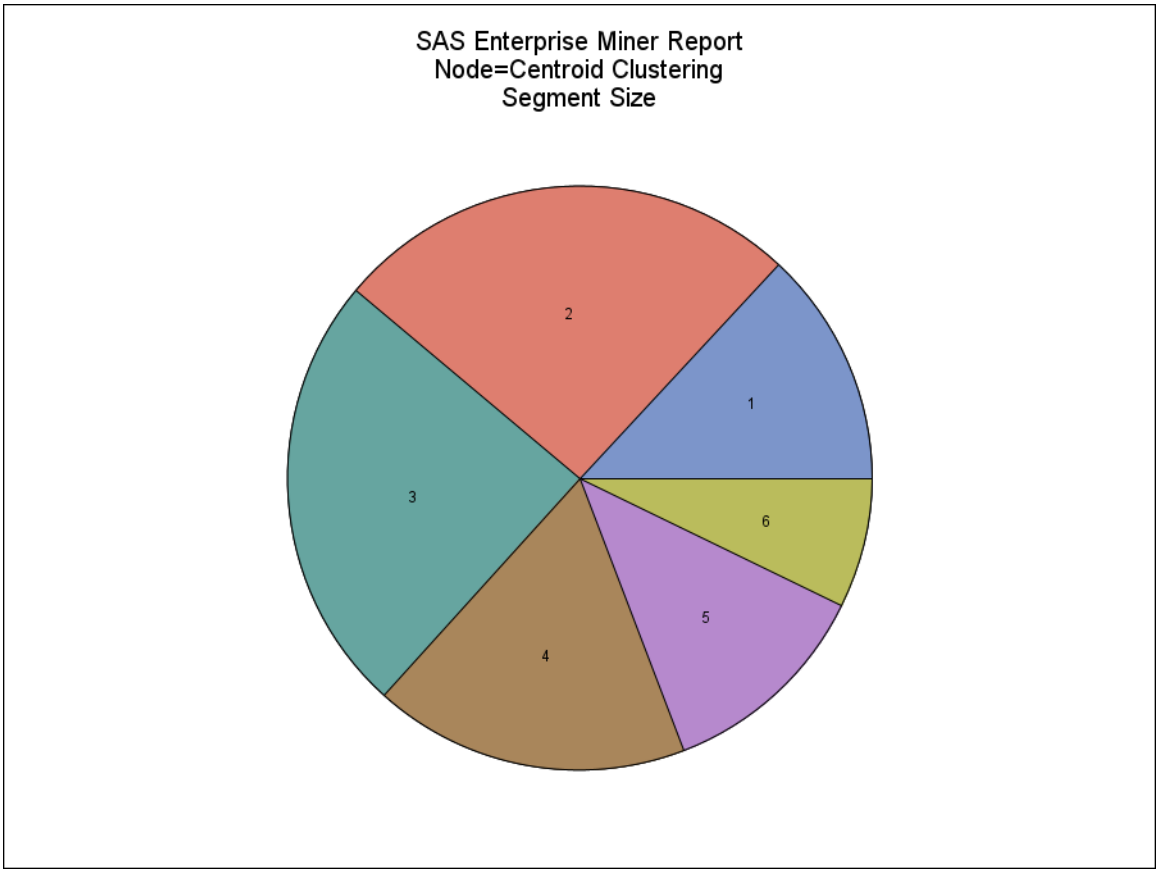
Node=Centroid Clustering Variable Summary

Role	Level	Frequency Count	Name
INPUT	INTERVAL	9	households housing_median_age latitude longitude median_house_value median_income population total_bedrooms total_rooms
INPUT	NOMINAL	1	ocean_proximity

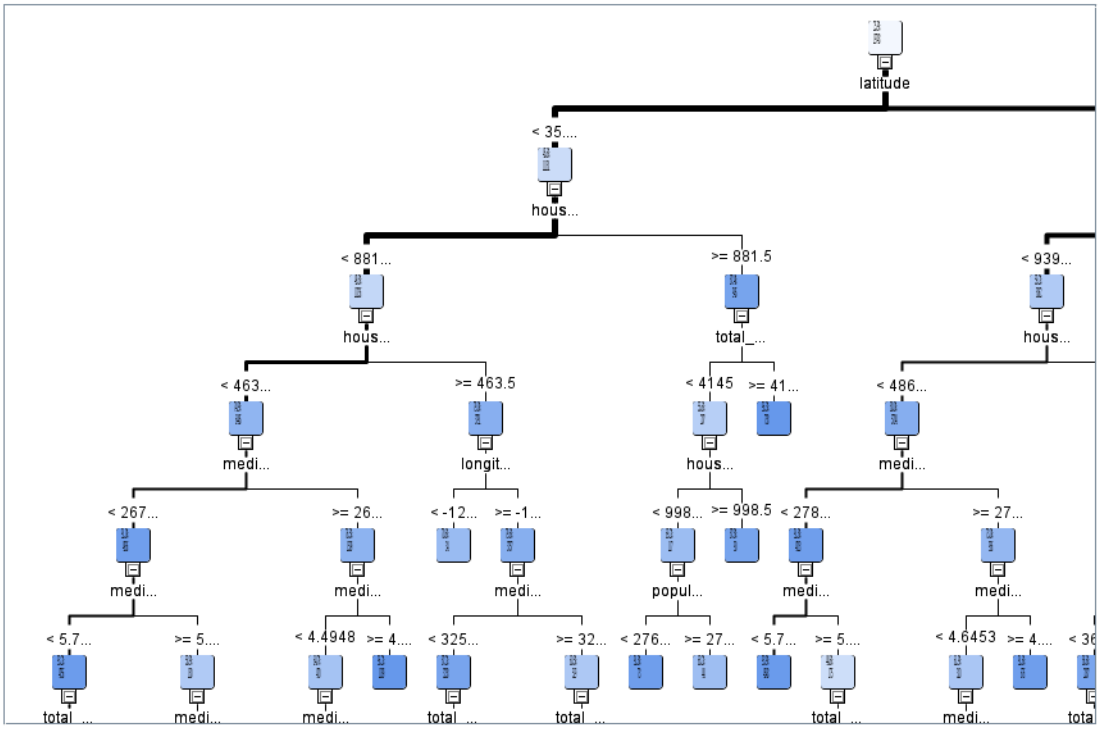
Node=Centroid Clustering Mean Statistics

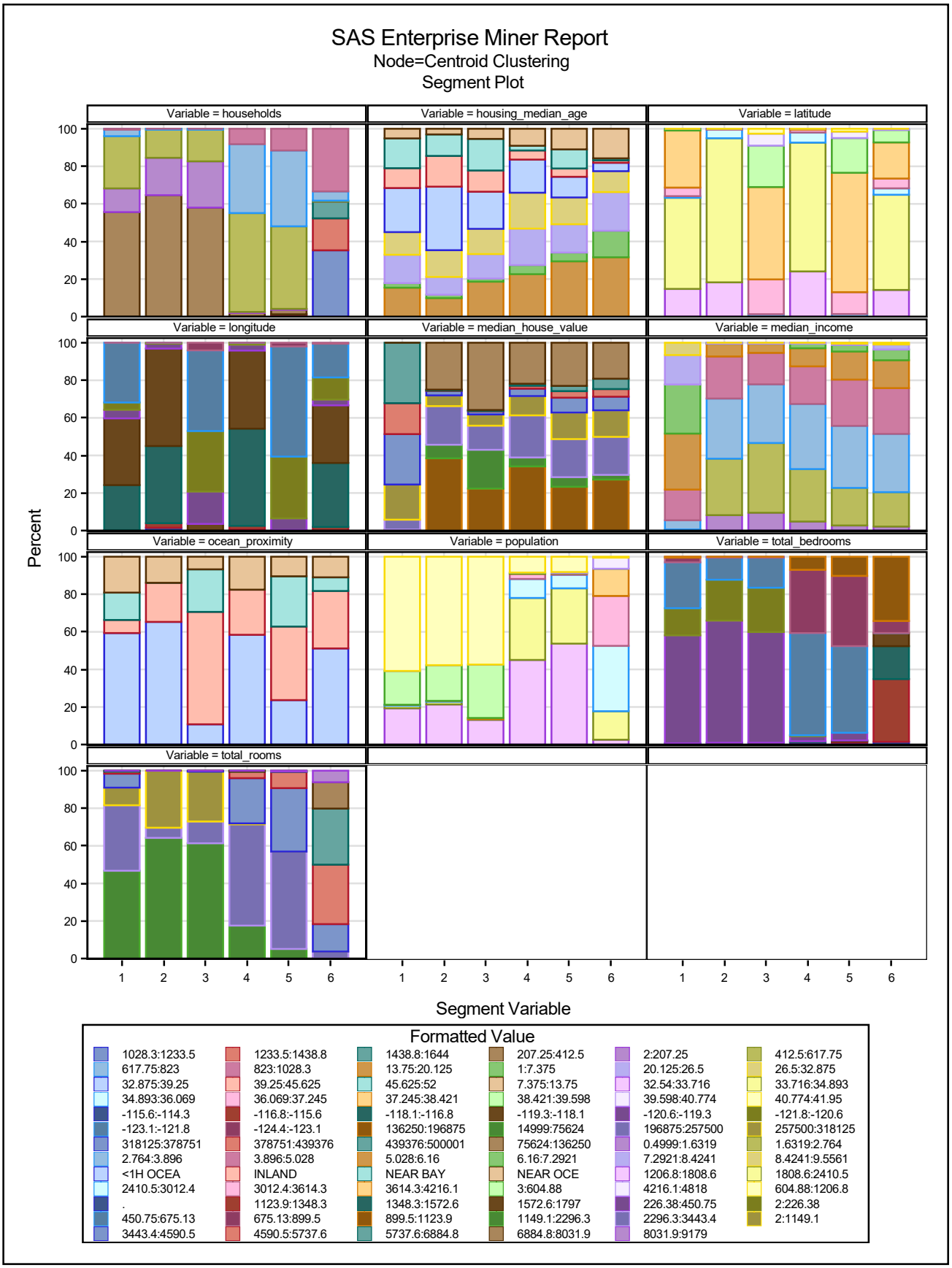
Variable	Segment 1	Segment 2	Segment 3	Segment 4	Segment 5	Segment 6
Clustering Criterion	0.60	0.60	0.60	0.60	0.60	0.60
Maximum Relative Change in Cluster Seeds	0.01	0.01	0.01	0.01	0.01	0.01
Improvement in Clustering Criterion	.	.	.	.	.	.
Frequency of Cluster	2586.00	5106.00	4792.00	3497.00	2359.00	1403.00
Root-Mean-Square Standard Deviation	0.69	0.50	0.58	0.57	0.64	0.77
Maximum Distance from Cluster Seed	5.04	4.27	5.22	4.61	5.25	5.51
Nearest Cluster	2.00	4.00	5.00	2.00	3.00	4.00
Distance to Nearest Cluster	2.96	2.46	2.74	2.46	2.74	3.75
households	355.69	298.41	292.30	619.66	646.59	1110.82
housing_median_age	31.36	33.30	31.26	24.78	25.13	18.50
latitude	35.22	33.90	38.03	33.92	37.93	35.18
longitude	-119.62	-117.97	-121.50	-117.96	-121.82	-119.16
median_house_value	385426.55	167948.69	140105.13	187661.23	207806.97	222046.29
median_income	6.15	3.22	3.01	3.45	3.83	4.03

Variable	Segment 1	Segment 2	Segment 3	Segment 4	Segment 5	Segment 6
population	930.74	938.88	821.70	1832.69	1739.93	3033.01
total_bedrooms	371.76	314.66	319.45	668.87	692.78	1201.51
total_rooms	2230.01	1424.32	1537.28	3031.05	3428.03	5793.53
ocean_proximity=<1H OCEA	0.59	0.65	0.11	0.58	0.24	0.51
ocean_proximity=INLAND	0.07	0.21	0.60	0.24	0.39	0.31
ocean_proximity=NEAR BAY	0.15	0.00	0.23	0.00	0.27	0.07
ocean_proximity=NEAR OCE	0.19	0.14	0.07	0.18	0.10	0.11



SAS Enterprise Miner Report
Node=Centroid Clustering
Tree Diagram




SAS Enterprise Miner Report
Node=Centroid Clustering
Segment Plot
Variable = households
Variable = housing_median_age
Variable = latitude
Variable = longitude
Variable = median_house_value
Variable = median_income
Variable = ocean_proximity
Variable = population
Variable = total_bedrooms
Variable = total_rooms
Percent
Segment Variable
Formatted Value
1028.3:1233.5
1233.5:1438.8
1438.8:1644
207.25:412.5
2:207.25
412.5:617.75
617.75:823
823:1028.3
13.75:20.125
1:7.375
20.125:26.5
26.5:32.875
32.875:39.25
39.25:45.625
45.625:52
7.375:13.75
32.54:33.716
33.716:34.893
34.893:36.069
36.069:37.245
37.245:38.421
38.421:39.598
39.598:40.774
40.774:41.95
-115.6:-114.3
-116.8:-115.6
-118.1:-116.8
-119.3:-118.1
-120.6:-119.3
-121.8:-120.6
-123.1:-121.8
-124.4:-123.1
136250:196875
14999:75624
196875:257500
257500:318125
318125:378751
378751:439376
439376:500001
75624:136250
0.4999:1.6319
1.6319:2.764
2.764:3.896
3.896:5.028
5.028:6.16
6.16:7.2921
7.2921:8.4241
8.4241:9.5561
<1H OCEA
INLAND
NEAR BAY
NEAR OCE
1206.8:1808.6
1808.6:2410.5
2410.5:3012.4
3012.4:3614.3
3614.3:4216.1
3:604.88
4216.1:4818
604.88:1206.8
.
1123.9:1348.3
1348.3:1572.6
1572.6:1797
226.38:450.75
2:226.38
450.75:675.13
675.13:899.5
899.5:1123.9
1149.1:2296.3
2296.3:3443.4
2:1149.1
3443.4:4590.5
4590.5:5737.6
5737.6:6884.8
6884.8:8031.9
8031.9:9179

SAS Enterprise Miner Report

Node=Ward Clustering
Summary

Node id = Clus
Node label = Ward Clustering
Meta path = Ids => Filter => Clus
Notes =

Node=Ward Clustering
Properties

Property	Value	Default	Property	Value	Default	Property	Value	Default
Component	Cluster		FinalMaxNum	20		MissingNominal	DEFAULT	
AutomaticMaxNum	50		HideVariable	Y		MissingOrdinal	DEFAULT	
AutomaticMethod	WARD		ImputationMethod	NONE		NominalEncoding	GLM	
AutomaticMinNum	2		Initial	DEFAULT		NumberClusterMethod	AUTOMATIC	
Bins	100		Learn	.		OrdinalEncoding	RANK	
CCCCutOff	3		LearnFinal	0.02		Radius	0	
ClusterGraphs	Y		LearnInitial	0.5		Stdize	STD	
Clusvar	_SEGMENT_		LearnSteps	1000		Summary	N	
ClusvarLabel	Segment Variable		MaxC	10		TrainDefaults	Y	
ClusvarRole	SEGMENT		MaxIter	10		TreeProfile	Y	
DistancePlot	Y		MaxSteps	1200		XConv	0.0001	
Drift	N		MissingInterval	DEFAULT				

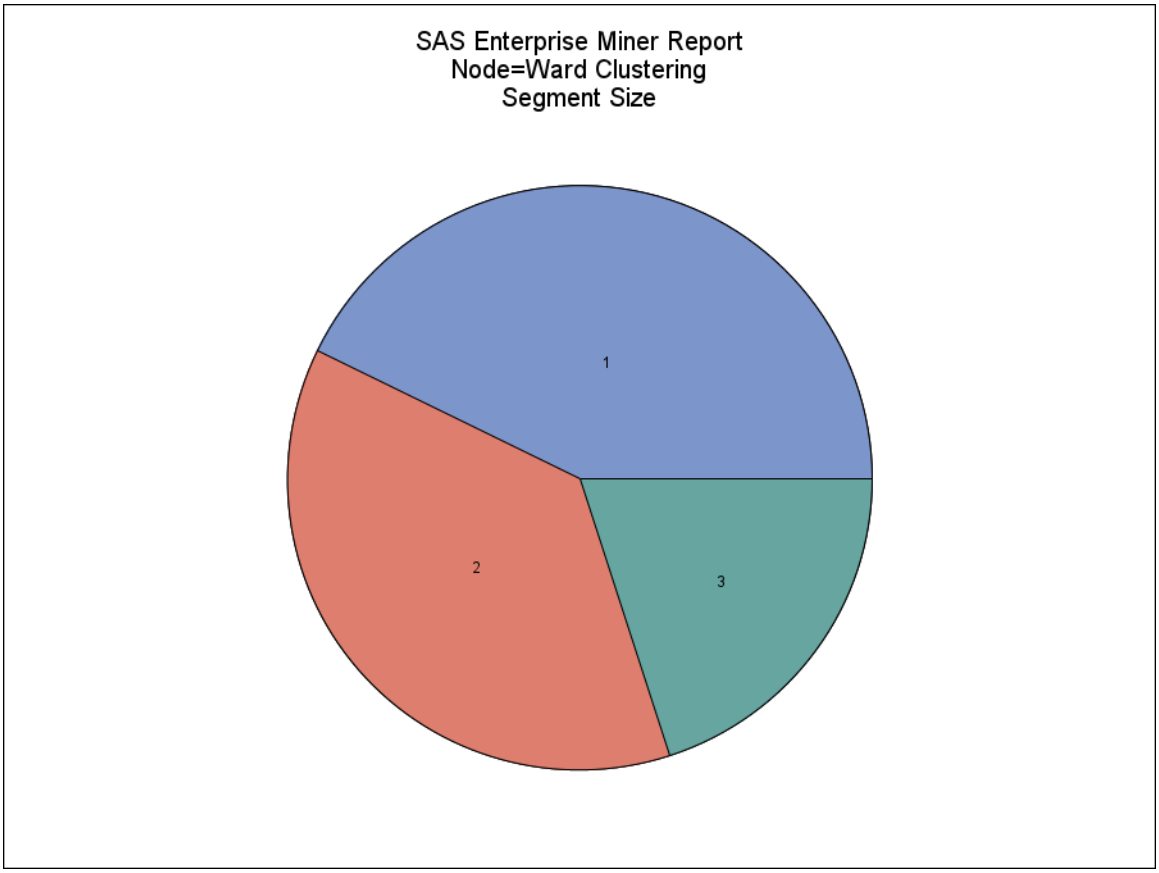
Node=Ward Clustering
Variable Summary

Role	Level	Frequency Count	Name
INPUT	INTERVAL	9	households housing_median_age latitude longitude median_house_value median_income population total_bedrooms total_rooms
INPUT	NOMINAL	1	ocean_proximity

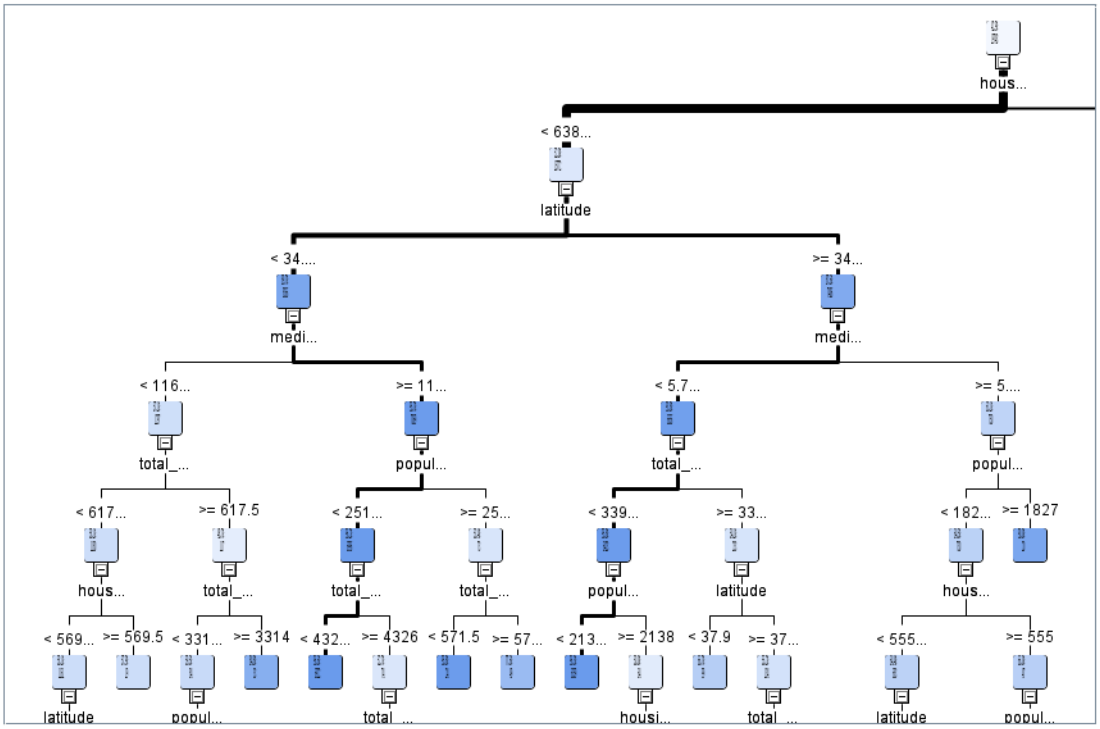
Node=Ward Clustering
Mean Statistics

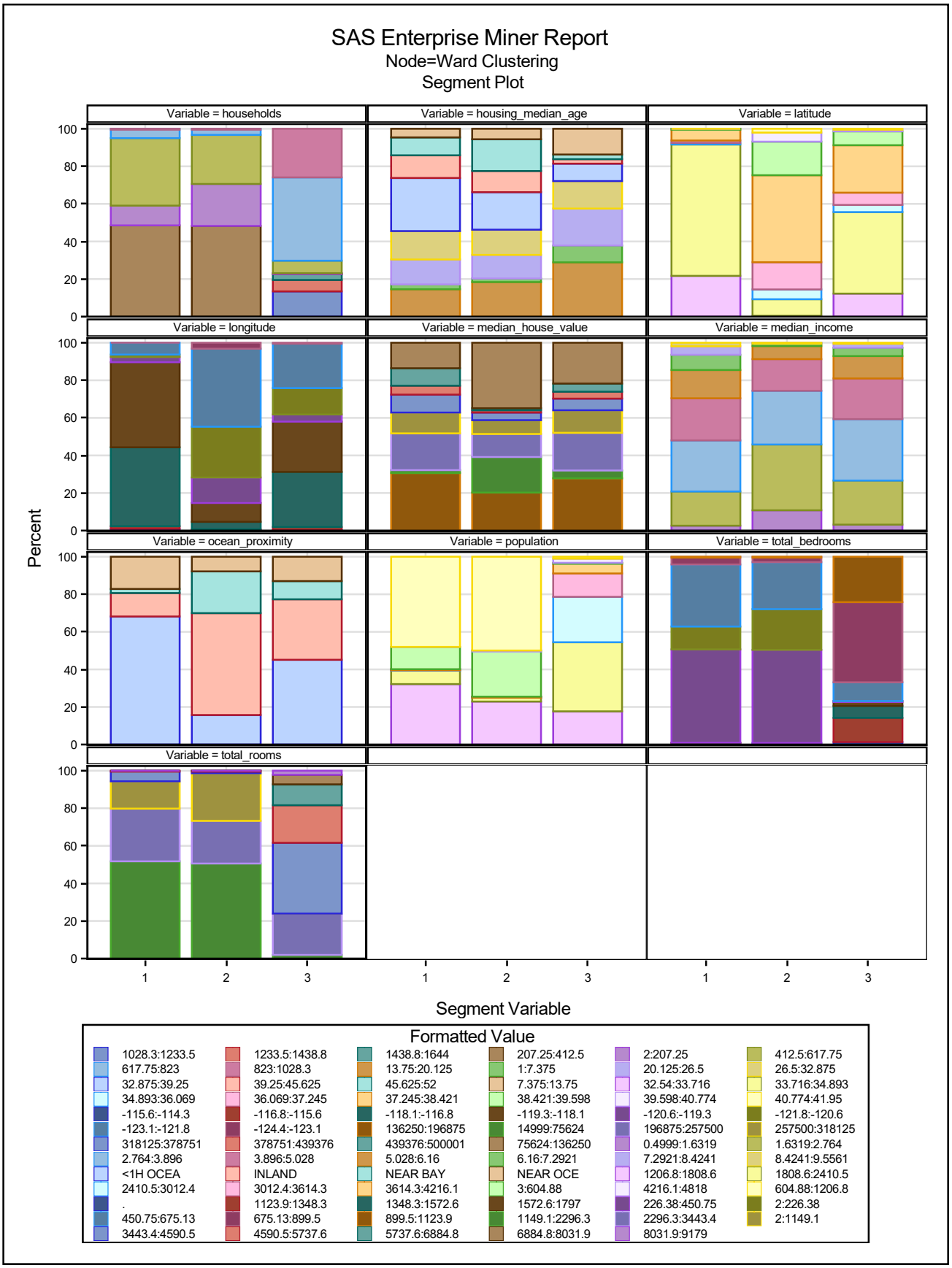
Variable	Segment 1	Segment 2	Segment 3
Clustering Criterion	0.70	0.70	0.70
Maximum Relative Change in Cluster Seeds	0.03	0.03	0.03
Improvement in Clustering Criterion	.	.	.
Frequency of Cluster	8466.00	7326.00	3951.00
Root-Mean-Square Standard Deviation	0.66	0.67	0.82
Maximum Distance from Cluster Seed	5.49	5.28	6.05
Nearest Cluster	2.00	1.00	1.00
Distance to Nearest Cluster	2.56	2.56	3.80
households	382.20	328.10	874.61
housing_median_age	30.69	31.25	21.29
latitude	34.09	37.53	35.53
longitude	-118.28	-121.14	-119.50
median_house_value	240529.88	153071.20	209039.79
median_income	4.24	3.10	3.78

Variable	Segment 1	Segment 2	Segment 3
population	1124.14	922.36	2438.09
total_bedrooms	403.33	355.94	945.64
total_rooms	2003.56	1710.16	4487.44
ocean_proximity=<1H OCEA	0.68	0.16	0.45
ocean_proximity=INLAND	0.12	0.54	0.32
ocean_proximity=NEAR BAY	0.02	0.22	0.10
ocean_proximity=NEAR OCE	0.17	0.08	0.13



SAS Enterprise Miner Report
Node=Ward Clustering
Tree Diagram




SAS Enterprise Miner Report
Node=Ward Clustering
Segment Plot
Variable = households
Variable = housing_median_age
Variable = latitude
Variable = longitude
Variable = median_house_value
Variable = median_income
Variable = ocean_proximity
Variable = population
Variable = total_bedrooms
Variable = total_rooms
Percent
0
20
40
60
80
100
1
2
3
Segment Variable
Formatted Value
1028.3:1233.5
1233.5:1438.8
1438.8:1644
207.25:412.5
2:207.25
412.5:617.75
617.75:823
823:1028.3
13.75:20.125
1:7.375
20.125:26.5
26.5:32.875
32.875:39.25
39.25:45.625
45.625:52
7.375:13.75
32.54:33.716
33.716:34.893
34.893:36.069
36.069:37.245
37.245:38.421
38.421:39.598
39.598:40.774
40.774:41.95
-115.6:-114.3
-116.8:-115.6
-118.1:-116.8
-119.3:-118.1
-120.6:-119.3
-121.8:-120.6
-123.1:-121.8
-124.4:-123.1
136250:196875
14999:75624
196875:257500
257500:318125
318125:378751
378751:439376
439376:500001
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3.896:5.028
5.028:6.16
6.16:7.2921
7.2921:8.4241
8.4241:9.5561
<1H OCEA
INLAND
NEAR BAY
NEAR OCE
1206.8:1808.6
1808.6:2410.5
2410.5:3012.4
3012.4:3614.3
3614.3:4216.1
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604.88:1206.8
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675.13:899.5
899.5:1123.9
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2296.3:3443.4
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5737.6:6884.8
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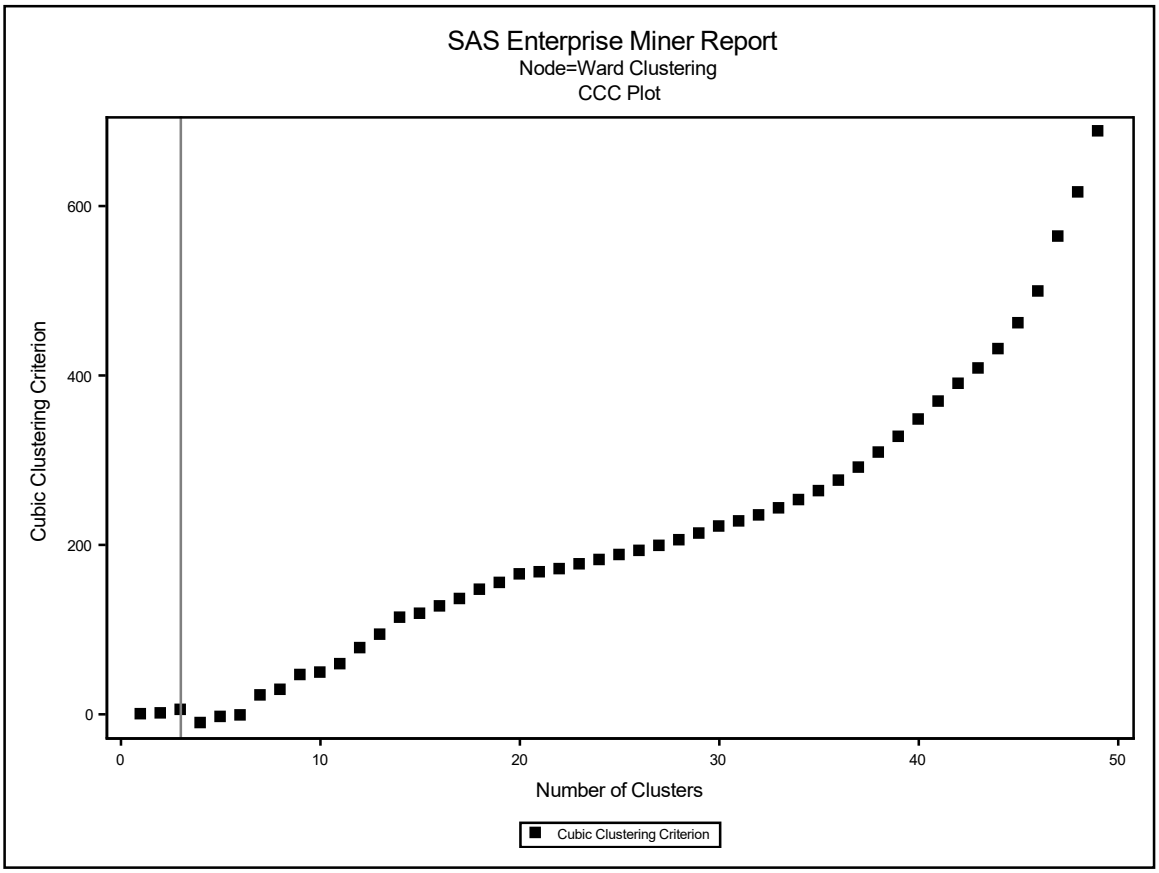

End of Report